JOE RANJITH L

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PROFESSIONAL SUMMARY

Pre-final year Electrical & Electronics Engineering student with hands-on experience in IoT system design, embedded programming, and real-time monitoring solutions. Built and field-deployed projects including a low-cost soil nutrient analyser, industrial energy & water monitoring system, and a smart gas detection system. Proficient in ESP8266/ESP32, Arduino, and cloud-integrated IoT applications, with additional exposure to electrical MEP design using AutoCAD and Revit. Seeking roles in IoT, embedded systems, or software development.

TECHNICAL SKILLS

Programming: C, C++, Python, Arduino (Embedded C), Data Structures, OOP
IoT & Embedded Systems: ESP8266, ESP32, Sensor Integration (MQ-2, NPK), Real-Time Data Acquisition, Circuit Design & Prototyping
Cloud & Connectivity: Firebase Realtime Database, ThingSpeak, Blynk, REST APIs
Industrial & Design Tools: PLC Programming, SCADA Systems, PSCAD Simulation, AutoCAD (Electrical), Revit
Tools & Technologies: 3D Printing, CAD Modelling, Git & GitHub, Basic Web Development

PROJECTS

Low-Cost Soil Nutrient Analyser *2024 – 2025*

- Developed an NPK sensor-based system under ₹3,000 for affordable real-field soil testing
- Built a Firebase-backed web app providing real-time soil monitoring and fertilizer recommendations
- Designed a 3D-printed, weather-resistant enclosure for field deployment
- Presented at Smart India Hackathon, Vel Tech Expo, and CMR Project Expo; Won 2nd Prize at Vel Tech High Tech Project Expo (50+ teams); Showcased at TANCAM TN Impact Program 2026

Energy & Water Monitoring System — TANCAM *2026*

- Built a low-cost IoT system for real-time energy & water usage monitoring in MSME industries
- Used ESP32, PZEM-004T, and YF-S201 sensors to measure voltage, current, power, energy, and flow rate
- Deployed a web-based dashboard for live monitoring, improving resource efficiency and fault awareness

IoT-Based Gas & Smoke Detection System *Jan – Mar 2024*

- Built real-time LPG leak and smoke detection using MQ-2 sensor with ESP8266 for instant mobile alerts
- Added buzzer and LED safety indicators; optimized for continuous low-power monitoring

EDUCATION

B.E. Electrical & Electronics Engineering 2023 – 2027
S.A. Engineering College, Chennai | CGPA: 7.60
Higher Secondary Certificate (HSC) 2023
Everwin Matriculation School, Chennai | Percentage: 70%

CERTIFICATIONS & ACHIEVEMENTS

Certifications: IoT | PLC | SCADA | PSCAD

- 2nd Prize – Vel Tech High Tech Project Expo 2025 (among 50+ competing teams)
- Smart India Hackathon 2024 – National-level participant
- CMR Project Expo 2025, Hyderabad – Project Presenter
- Active Member – AI Club, S.A. Engineering College